

Extending PRCTL with conditional expected reward operators

Eshita Zaman¹[0009–0008–5680–5509], Gianfranco Ciardo²[0000–0002–4906–6145],
and Andrew S. Miner²[0000–0002–7737–6888]

¹ Department of Computer Science, Utah Valley University, Orem, UT, USA
`eshita.zaman@uvu.edu`

² Department of Computer Science, Iowa State University, Ames, IA, USA
`{ciardo, asminer}@iastate.edu`

Abstract. Probabilistic model checking verifies systems where events occur probabilistically, as opposed to nondeterministically. Most probabilistic temporal logics express the probability of paths satisfying some property, possibly within a time interval, in a discrete-time (PCTL) or continuous-time (CSL) Markov chain; some include the idea of rewards (PRCTL). We extend PRCTL with operators for expected accumulated reward, reward rate, and impulse reward up to (1) satisfying a path formula or (2) a given time t unrelated to when the path formula becomes satisfied, but conditioned on it being satisfied. These cannot be expressed by prior probabilistic reward logics. We provide efficient algorithms to evaluate these operators and results from our prototype implementation. Keywords: Markov reward models, expected rewards, PRCTL.

1 Introduction

Model checking can verify logical properties in models of complex hardware and software systems, but fault-tolerant systems, computer networks, and protocols are stochastic and require quantitative probabilistic analysis, not just the qualitative analysis of classic model checking [8, 25]. Example properties are “the chance of failure within the first 1,000 hours is at most 0.01%” or “in the long run, the system is in a working state 98% of the time”. Temporal logics PCTL [21] and CSL [6] are probabilistic analogues to CTL [16], respectively defined over discrete-time Markov chains (DTMCs) and continuous-time Markov chains (CTMCs). These logics, as well as PCTL* [4], asCSL [5], and CSL^{TA} [19] require tools to verify the probabilistic properties they can express. PRISM [28], SMART [14], and Storm [18] are examples of such tools developed in academia.

Probabilistic model checking studies transient and steady-state (long-run) system behavior, but some questions cannot be expressed by classical probabilistic temporal logics, which have thus been extended in the last few decades. [2] extended PCTL with an operator $\mathbb{S}$ to query the long-run behavior of DTMCs (the continuous-time counterpart of $\mathbb{S}$ had already been defined in CSL) and proposed probabilistic reward computation tree logic (PRCTL) to reason about performance measures of systems modeled as DMRMs (discrete-time Markov reward models), but only provided an algorithm to compute state-based rewards. [7] proposed the reward-based logic CRL for CTMCs.

[27] presented efficient algorithms to compute expected accumulated reward (up to a time t or upon reaching a condition) and expected instantaneous reward (at time t), and mentioned (without details) that similar algorithms can compute the expected reward for “until” (path) formulas; unlike [2], it considered (state) rate rewards and (transition) impulse rewards, in DTMCs and CTMCs. However, [2, 27] addressed expected rewards only for until formulas with time intervals having zero lower bound. In 1978, [12] showed how to compute the expected reward up to absorption as the time to absorption of a related process, but the approach was limited to absorbing Markov chains with unbounded time intervals.

In contrast, we show how to compute conditional expected rewards for time-unbounded and time-bounded until formulas with arbitrary time bounds. Our extensions include efficient computation of the conditional expected rewards up to satisfying a PRCTL path formula φ or up to a given time t unrelated to the time φ becomes satisfied on each path. Our logic can answer more general queries regarding expected rewards, including the expected time to satisfy a path formula (the special case when all state reward rates are 1).

2 Preliminaries

We consider probabilistic systems whose underlying stochastic model is a (labeled) *discrete-time Markov chain* (DTMC) with finite state space [24].

Definition 1. A discrete-time discrete-state stochastic process $\{X^{[k]} : k \in \mathbb{N}\}$ describing the state $X^{[k]} \in \mathcal{S}$ at time k is a (homogeneous) DTMC with discrete state space $\mathcal{S}$ iff $\Pr(X^{[k+1]} = j | X^{[k]} = i_k \wedge \dots \wedge X^{[0]} = i_0) = \Pr(X^{[k+1]} = j | X^{[k]} = i_k)$. Such DTMC is specified by a tuple $\mathcal{M} = (\mathcal{S}, \mathbf{P}, \mathcal{A}, L)$ where:

- $\mathcal{S}$ is a nonempty discrete set of *states*, which we assume to be finite;
- $\mathbf{P} : \mathcal{S} \times \mathcal{S} \rightarrow [0, 1]$, $\forall i \in \mathcal{S}, \sum_{j \in \mathcal{S}} \mathbf{P}[i, j] = 1$, is the *transition probability matrix*, i.e., for any $k \in \mathbb{N}$, $\Pr(X^{[k+1]} = j \mid X^{[k]} = i) = \mathbf{P}[i, j]$;
- $\mathcal{A}$ is a set of *atomic propositions*;
- $L : \mathcal{S} \rightarrow 2^{\mathcal{A}}$ specifies the atomic propositions holding in each state. $\square$

Matrix $\mathbf{P}$ implies a directed graph with node set $\mathcal{S}$ and edge set corresponding to its positive entries.

Definition 2. A discrete-time Markov reward model (DMRM) is $\mathcal{D} = (\mathcal{M}, \rho)$, where $\mathcal{M} = (\mathcal{S}, \mathbf{P}, \mathcal{A}, L)$ is a DTMC and ρ is a *reward structure*: $\rho[i] \in [0, \infty)$ is the rate at which $\mathcal{M}$ gains reward while in state $i \in \mathcal{S}$, while $\rho[i, j] \in [0, \infty)$ is the impulse reward $\mathcal{M}$ gains when transitioning from state $i \in \mathcal{S}$ to state $j \in \mathcal{S}$. $\square$

Either the reward rate vector $\rho[\mathcal{S}]$ or the reward impulse matrix $\rho[\mathcal{S}, \mathcal{S}]$ can be omitted (i.e., be uniformly zero); if both are zero, the DMRM is a DTMC. In general, one can specify multiple reward structures $\rho_1, \dots, \rho_n$ [2] over a given DMRM. From now on, we only use the term DMRM, understanding that we are referring to just the DTMC whenever the reward structure is irrelevant.

Definition 3. An infinite path in a DMRM starting from state i_0 is a sequence of states $\sigma = (i_0, i_1, i_2, \dots) \in \mathcal{S}^\omega$, where $\mathbf{P}[i_k, i_{k+1}] > 0$, for $k \geq 0$. Given any $k \in \mathbb{N}$, let $\sigma[k]$ be the state i_k after the k^{th} transition and $\sigma[0, k]$ be the finite prefix $= (i_0, \dots, i_k) \in \mathcal{S}^{k+1}$ of σ describing a path of length k . $\square$

Definition 4. State j is *reachable* from state i , written $i \rightarrow j$, iff the DMRM has a path from i to j . Then, state i is *recurrent* if $\forall j \in \mathcal{S}, i \rightarrow j$ implies $j \rightarrow i$, otherwise state i is *transient*. If $\mathbf{P}[i, i] = 1$, state i is *absorbing*; this is a special case of a recurrent state: if the DMRM enters state i , it stays in i forever. $\square$

Reachability between states can be determined by graph analysis, considering only whether the specific transition probability values are zero or positive. States i and j are mutually reachable, $i \leftrightarrow j$, if $i \rightarrow j$ and $j \rightarrow i$. Mutual reachability is an equivalence relation that partitions the state space $\mathcal{S}$ into classes with the same properties: if a state is recurrent (or transient), all the states in its class are recurrent (or transient). In graph terms, the equivalence classes are the strongly connected components (SCCs) of the graph. SCCs with no edges to states of another SCC (i.e., *terminal* SCCs) correspond to the recurrent classes.

For PCTL model checking, we must consider only *fair paths* [9] in the DTMC. In our case, this means either finite paths or infinite paths that do not remain in non-terminal SCCs forever, but instead eventually reach a terminal SCC and visit each of its states infinitely many times. Unfair paths have probability measure zero and can be excluded from the computation through graph analysis. This is relevant only for time-unbounded path formulas since, for a finite upper time bound k , all paths are fair so we consider all paths of length up to k .

Definition 5. A DMRM is *irreducible* iff it contains only one class, meaning that all its states are mutually reachable, thus recurrent. $\square$

Definition 6. A DMRM with initial distribution $\pi^{[0]}$, i.e., $\forall j \in \mathcal{S}, \Pr(X^{[0]} = j) = \pi^{[0]}[j]$, has *limiting distribution* π if $\forall j \in \mathcal{S}, \lim_{k \rightarrow \infty} \Pr(X^{[k]} = j) = \pi[j]$. $\square$

The limiting distribution π always exists unless the DMRM is *periodic* [24]; however, the following definition extends this concept to periodic DMRMs.

Definition 7. A DMRM with initial distribution $\pi^{[0]}$ always has a *Cesàro mean distribution* $\bar{\pi}$, defined as the limit of the average of the state probability vectors over time: $\forall j \in \mathcal{S}, \bar{\pi}[j] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \Pr(X^{[k]} = j)$. $\square$

Definition 8. Given finite path $\sigma = (i_0, \dots, i_k) \in \mathcal{S}^{k+1}$, its accumulated reward is $\rho(\sigma) = \sum_{n=0}^{k-1} (\rho[i_n] + \rho[i_n, i_{n+1}])$. $\square$

Thus, for the expected accumulated reward *up to* time k , we consider impulse rewards until k^+ (including the impulse reward gained upon entering state i_k at time k) but reward rates until k^- (the reward rate of state i_k is irrelevant). For the expected reward rate *at* time k , we consider the reward rate at time k^+ . For the expected impulse reward *at* time k , we consider the impulse reward earned by the transition at time k if $k \in \mathbb{N}^{\geq 1}$ (the impulse reward at time $k=0$ is 0).

Definition 9. Let $Paths(i_0) \subseteq \mathcal{S}^\omega$, and $Paths(i_0, k) \subseteq \mathcal{S}^{k+1}$, denote the set of infinite fair paths, and length- k paths, starting from state i_0 . The set of paths with prefix $(i_0, \dots, i_k)$ has probability $\Pr(i_0, \dots, i_k) = \mathbf{P}[i_0, i_1] \cdots \mathbf{P}[i_{k-1}, i_k]$. Then:

(1) The expected reward rate at time k^+ , starting from state i_0 , is:

$E_R(i_0, k) = \sum_{j \in \mathcal{S}} \mathbf{P}^k[i_0, j] \cdot \rho[j]$, where $\mathbf{P}^k[i_0, j]$ is the probability that, starting in state i_0 , the DMRM is in state j after k steps. Let $\mathbf{P}^\infty[i_0, j] = \lim_{k \rightarrow \infty} \mathbf{P}^k[i_0, j]$, be the limiting distribution (in the Cesàro sense, if the DMRM is periodic).

(2) The expected impulse reward at time k starting from state i_0 is:

$$E_I(i_0, k) = \sum_{j \in \mathcal{S}} \mathbf{P}^{k-1}[i_0, j] \cdot \sum_{j' \in \mathcal{S}} \mathbf{P}[j, j'] \cdot \rho[j, j'].$$

(3) The expected accumulated reward up to time k starting from state i_0 is:

$$E_A(i_0, k) = \sum_{j \in \mathcal{S}} \sum_{n=0}^{k-1} \mathbf{P}^n[i_0, j] \left(\rho[j] + \sum_{j' \in \mathcal{S}} \mathbf{P}[j, j'] \cdot \rho[j, j'] \right). \quad \square$$

3 PRCTL and proposed extensions

[2] extended the logic PCTL [21] to express performance properties in a DMRM. It proposed four new operators: $\mathbb{S}$ to compute the long-run average, $\mathcal{E}$, $\mathcal{C}$, and $\mathcal{Y}$ to express reward measures along with the original PCTL operators, resulting in the *probabilistic reward computation tree logic* (PRCTL). The syntax of PRCTL for a *state property* ψ and a *path property* φ is, respectively:

$$\begin{aligned} \psi &::= \text{true} \mid a \mid \psi \wedge \psi \mid \neg \psi \mid \mathbb{P}_{\bowtie z}(\varphi) \mid \mathbb{S}_{\bowtie z}(\psi) \mid \mathcal{E}_{\bowtie r}^t(\psi) \mid \mathcal{C}_{\bowtie r}^t(\psi) \mid \mathcal{Y}_{\bowtie r}^t(\psi) \\ \varphi &::= \mathbf{X}\psi \mid \psi \mathbf{U}^{[h, k]} \psi \mid \psi \mathbf{U}^{[h, \infty)} \psi, \end{aligned}$$

where $a \in \mathcal{A}$, $\bowtie \in \{<, \leq, =, >, \geq\}$, $z \in [0, 1]$, and $r \in [0, \infty]$ is a reward value.

The semantics of these operators, assuming i_0 is the starting state at time 0, is:

$i_0 \models \mathcal{E}_{\bowtie r}^t(\psi)$, i.e., i_0 satisfies $\mathcal{E}_{\bowtie r}^t(\psi)$, if the expected reward rate of states satisfying ψ on paths of length t starting at i_0 is $\bowtie r$:

$$\sum_{\sigma=(i_0, \dots, i_t)} (\Pr(\sigma) \cdot \sum_{0 \leq k < t, i_k \models \psi} \rho[i_k]/t) \bowtie r.$$

$i_0 \models \mathcal{E}_{\bowtie r}^\infty(\psi)$, i.e., in the time-unbounded case, if

$$\sum_{j \models \psi} \mathbf{P}^\infty[i_0, j] \cdot \rho[j] \bowtie r.$$

$i_0 \models \mathcal{C}_{\bowtie r}^t(\psi)$ if the expected reward rate at time t (possibly ∞) in ψ states is $\bowtie r$:

$$\sum_{j \models \psi} \mathbf{P}^t[i_0, j] \cdot \rho[j] \bowtie r.$$

$i_0 \models \mathcal{Y}_{\bowtie r}^t(\psi)$ if the expected accumulated reward up to time t in ψ states is $\bowtie r$:

$$\sum_{\sigma=(i_0, \dots, i_t)} (\Pr(\sigma) \cdot \sum_{0 \leq k < t, i_k \models \psi} \rho[i_k]) \bowtie r$$

$i_0 \models \mathcal{Y}_{\infty}^\infty(\psi)$ if i_0 can reach a recurrent class containing a state with positive reward rate; otherwise, $i_0 \models \mathcal{Y}_{\bowtie r}^\infty(\psi)$ if, letting $\mathcal{R}$ be the recurrent states,

$$\sum_{\sigma=(i_0, \dots, i_t), i_0, \dots, i_{t-1} \notin \mathcal{R}, i_t \in \mathcal{R}} (\Pr(\sigma) \cdot \sum_{0 \leq k < t, i_k \models \psi} \rho[i_k]) \bowtie r,$$

$\llbracket \text{true} \rrbracket$	$= \mathcal{S}$	Standard PRCTL judgment rules
$\llbracket a \rrbracket$	$= \{i \mid a \in L(i)\}$	(where $\llbracket \psi \rrbracket = \{i \in \mathcal{S} \mid i \models \psi\}$ and $\llbracket \varphi \rrbracket_i = \{\sigma \in \text{Paths}(i) \mid \sigma \models \varphi\}$)
$\llbracket \psi_1 \wedge \psi_2 \rrbracket$	$= \llbracket \psi_1 \rrbracket \cap \llbracket \psi_2 \rrbracket$	
$\llbracket \neg \psi \rrbracket$	$= \mathcal{S} \setminus \llbracket \psi \rrbracket$	
$\llbracket \mathbb{P}_{\bowtie z}(\varphi) \rrbracket$	$= \{i_0 \mid \Pr(\varphi, i_0) \bowtie z\}$	
$\llbracket \mathbb{S}_{\bowtie z}(\psi) \rrbracket$	$= \{i_0 \mid \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \Pr\{\sigma[k] \in \psi \mid \sigma \in \text{Paths}(i_0)\} \bowtie z\}$	
$\llbracket \mathbb{X}\psi \rrbracket_{i_0}$	$= \{\sigma \in \text{Paths}(i_0) \mid \sigma[1] \in \llbracket \psi \rrbracket\}$	
$\llbracket \psi_1 \mathbb{U}^{[h,k]} \psi_2 \rrbracket_{i_0}$	$= \{\sigma \in \text{Paths}(i_0) \mid \exists l \in [h, k], \sigma[l] \in \llbracket \psi_2 \rrbracket \wedge \forall l' \in [0, \dots, l), \sigma[l'] \in \llbracket \psi_1 \rrbracket\}$	
$\llbracket \psi_1 \mathbb{U}^{[h,\infty)} \psi_2 \rrbracket_{i_0}$	$= \{\sigma \in \text{Paths}(i_0) \mid \exists l \geq h, \sigma[l] \in \llbracket \psi_2 \rrbracket \wedge \forall l' \in [0, \dots, l), \sigma[l'] \in \llbracket \psi_1 \rrbracket\}$	
$\mathbb{C}^A(\varphi)_{i_0}$	$= \sum_{(i_0:i_u) \models \varphi} \Pr(i_0:i_u) \cdot \rho(i_0:i_u) / \mathbb{P}(\varphi)_{i_0}$	New PRCTL judgment rules
$\llbracket \mathbb{C}_{\bowtie r}^A(\varphi) \rrbracket$	$= \{i_0 \mid \mathbb{C}^A(\varphi)_{i_0} \bowtie r\}$	(assuming $\mathbb{P}(\varphi)_{i_0} > 0$, else the value is $\perp$)
$\mathbb{C}^R(\varphi)_{i_0}$	$= \sum_{(i_0:i_u) \models \varphi} \Pr(i_0:i_u) \cdot \rho[i_u] / \mathbb{P}(\varphi)_{i_0}$	
$\llbracket \mathbb{C}_{\bowtie r}^R(\varphi) \rrbracket$	$= \{i_0 \mid \mathbb{C}^R(\varphi)_{i_0} \bowtie r\}$	
$\mathbb{C}^I(\varphi)_{i_0}$	$= \sum_{(i_0:i_u) \models \varphi} \Pr(i_0:i_u) \cdot \rho[i_{u-1}, i_u] / \mathbb{P}(\varphi)_{i_0}$	
$\llbracket \mathbb{C}_{\bowtie r}^I(\varphi) \rrbracket$	$= \{i_0 \mid \mathbb{C}^I(\varphi)_{i_0} \bowtie r\}$	
$\mathbb{T}^A(\varphi, t)_{i_0}$	$= \begin{cases} \sum_{(i_0:i_u) \models \varphi, v=\max(t,u)} \Pr(i_0:i_v) \cdot \rho(i_0:i_t) / \mathbb{P}(\varphi)_{i_0} & \text{if } t < \infty \\ \infty & \text{if } t = \infty, \exists \sigma: \sigma[i_0:j] \models \varphi, j \in \mathcal{R}, \rho[j] > 0 \vee \exists j', \rho[j, j'] > 0 \\ \sum_{(i_0:i_u) \models \varphi} \Pr(i_0:i_u) \cdot \sum_{j \in \mathcal{R}} \Pr(i_u:j) \cdot \rho(i_0:i_u:j) / \mathbb{P}(\varphi)_{i_0} & \text{o.w.} \end{cases}$	
$\llbracket \mathbb{T}_{\bowtie r}^A(\varphi, t) \rrbracket$	$= \{i_0 \mid \mathbb{T}^A(\varphi, t)_{i_0} \bowtie r\}$	
$\mathbb{T}^R(\varphi, t)_{i_0}$	$= \begin{cases} \sum_{(i_0:i_u) \models \varphi, v=\max(t,u)} \Pr(i_0:i_v) \cdot \rho[i_t] / \mathbb{P}(\varphi)_{i_0} & \text{if } t < \infty \\ \sum_{(i_0:i_u) \models \varphi} \Pr(i_0:i_u) \cdot \sum_{j \in \mathcal{R}} \Pr(i_u:j) \cdot \pi(j \mathcal{B}_j) \cdot \rho[j] / \mathbb{P}(\varphi)_{i_0} & \text{o.w.} \end{cases}$	
$\llbracket \mathbb{T}_{\bowtie r}^R(\varphi, t) \rrbracket$	$= \{i_0 \mid \mathbb{T}^R(\varphi, t)_{i_0} \bowtie r\}$	
$\mathbb{T}^I(\varphi, t)_{i_0}$	$= \begin{cases} \sum_{(i_0:i_u) \models \varphi, v=\max(t,u)} \Pr(i_0:i_v) \cdot \rho[i_{t-1}, i_t] / \mathbb{P}(\varphi)_{i_0} & \text{if } t < \infty \\ \sum_{(i_0:i_u) \models \varphi} \Pr(i_0:i_u) \cdot \sum_{j, j' \in \mathcal{R}} \Pr(i_u:j, j') \cdot \pi(j \mathcal{B}_j) \cdot \rho[j, j'] / \mathbb{P}(\varphi)_{i_0} & \text{o.w.} \end{cases}$	
$\llbracket \mathbb{T}_{\bowtie r}^I(\varphi, t) \rrbracket$	$= \{i_0 \mid \mathbb{T}^I(\varphi, t)_{i_0} \bowtie r\}$	

Fig. 1: Judgment rules for standard (top) and new (bottom) PRCTL formulas. $(i_0:i_u)$ is a shorthand for the finite path $(i_0, \dots, i_u)$. If $\varphi = a\mathbb{U}^{[h,\infty)}b$, then u is the first time $\geq h$ that b is reached on a path of a states starting from i_0 , i.e., $i_0, \dots, i_{h-1} \models a$, $i_h, \dots, i_{u-1} \models a \wedge \bar{b}$, and $i_u \models b$; if $\varphi = a\mathbb{U}^{[h,k]}b$, then, in addition, $u \leq k$. For “ $\mathbb{T}$ ” formulas, $v = \max(t, u)$. $\mathcal{R}$ is the recurrent states and $\mathcal{B}_j$ is the recurrent class containing state j .

Figure 1 (top) presents the semantics judgment rules of PRCTL (note that, in [2], operators $\mathbb{S}$, $\mathcal{E}$, $\mathcal{C}$, and $\mathcal{V}$ apply only to a PRCTL state formula ψ , not to a path formula φ). [27] provided a new operator $\mathbb{R}$ to compute expected rewards for both state properties and path properties. Section 6 shows how the operators defined in [2] and [27] can be expressed using our proposed extensions.

3.1 Proposed extensions to PRCTL

We modify and extend the PRCTL state formula syntax to express more general reward-based properties. The new syntax for a PRCTL state formula ψ is:

$$\psi ::= \text{true} \mid a \mid \psi \wedge \psi \mid \neg \psi \mid \mathbb{P}_{\bowtie z}(\varphi) \mid \mathbb{S}_{\bowtie z}(\psi) \mid \mathbb{C}_{\bowtie r}^A(\varphi) \mid \mathbb{C}_{\bowtie r}^R(\varphi) \mid \mathbb{C}_{\bowtie r}^I(\varphi) \mid \mathbb{T}_{\bowtie r}^A(\varphi, t) \mid \mathbb{T}_{\bowtie r}^R(\varphi, t) \mid \mathbb{T}_{\bowtie r}^I(\varphi, t),$$

where φ is a PRCTL path property (expressed in the syntax shown above). $\mathbb{C}^A$, $\mathbb{C}^R$, and $\mathbb{C}^I$ denote conditional expected accumulated reward, reward rate, and reward impulse; $\mathbb{T}^A$, $\mathbb{T}^R$, and $\mathbb{T}^I$ denote conditional expected accumulated reward up to time t , reward rate at time t , and reward impulse at time t , respectively.

We now define the semantics of the six new operators. As they describe a reward conditioned on starting in a state i_0 and satisfying a path formula φ , they all return the *undefined* value $\perp$ if *no* path leaving i_0 satisfies φ (this is detected using CTL model checking, without numerical computation). Fig. 1 (bottom) provides their semantics when at least one path leaving i_0 satisfies φ .

State i_0 satisfies $\mathbb{C}_{\bowtie r}^A(\varphi)$ if the expected accumulated reward, conditioned on paths from i_0 satisfying φ , is $\bowtie r$, where φ is a time-bounded or time-unbounded until formula (including $X\psi$ and $F\psi$, respectively equivalent to $\text{trueU}^{[1,1]}\psi$ and $\text{trueU}^{[0,\infty)}\psi$). State i_0 satisfies $\mathbb{C}_{\bowtie r}^R(\varphi)$, or $\mathbb{C}_{\bowtie r}^I(\varphi)$, if the expected reward rate, or impulse reward, at the earliest time φ is satisfied, conditioned on paths from i_0 satisfying φ , is $\bowtie r$. State i_0 satisfies $\mathbb{T}_{\bowtie r}^A(\varphi, t)$ if the expected accumulated reward until time t , conditioned on paths from i_0 satisfying φ , is $\bowtie r$. State i_0 satisfies $\mathbb{T}_{\bowtie r}^R(\varphi, t)$, or $\mathbb{T}_{\bowtie r}^I(\varphi, t)$, if the expected reward rate, or expected impulse reward, at time t , conditioned on paths from i_0 satisfying φ , is $\bowtie r$.

4 Model checking algorithms

4.1 Computing $\mathbb{S}$

We briefly recall how to compute the $\mathbb{S}$ operator. As discussed in [2], if the DTMC $\mathcal{M}$ underlying the DMRM is irreducible, its steady-state probabilities $\bar{\pi}$ (or π , if $\mathcal{M}$ is not periodic) can be determined by solving the linear system $\bar{\pi}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$, subject to $\sum_j \bar{\pi}[j] = 1$. If instead $\mathcal{M}$ contains only transient and absorbing states, it will eventually enter an absorbing state j , with a probability depending in general on the initial state i , $\bar{\pi}[j|i] = \lim_{k \rightarrow \infty} \Pr(X^{[k]} = j \mid X^{[0]} = i)$.

The general case is a DMRM with state space partitioned into transient states $\mathcal{S}^{(0)}$ and recurrent classes $\mathcal{S}^{(1)}, \dots, \mathcal{S}^{(n)}$. The limiting distribution again depends on the initial state i , i.e., $\bar{\pi}[\mathcal{S}|i] = [\bar{\pi}[\mathcal{S}^{(0)}|i], \bar{\pi}[\mathcal{S}^{(1)}|i], \dots, \bar{\pi}[\mathcal{S}^{(n)}|i]]$, with $\bar{\pi}[\mathcal{S}^{(0)}|i] = \mathbf{0}$ and $\bar{\pi}[\mathcal{S}^{(m)}|i] = \pi'[m|i] \cdot \bar{\gamma}^{(m)}$, where $\pi'[m|i]$ is the probability of reaching $\mathcal{S}^{(m)}$ starting from state i (computed on DMRM $\mathcal{M}'$ obtained from $\mathcal{M}$ by merging each recurrent class $\mathcal{S}^{(m)}$ into an absorbing state m), and $\bar{\gamma}^{(m)}$ is the Cesàro mean distribution for states in recurrent class $\mathcal{S}^{(m)}$ in isolation, computed from DTMC $\mathcal{M}^{(m)}$ with transition probability matrix $\mathbf{P}[\mathcal{S}^{(m)}, \mathcal{S}^{(m)}]$.

4.2 Computing $\mathbb{C}^A$, $\mathbb{C}^R$, and $\mathbb{C}^I$

We now present numerical methods to compute conditional rewards $\mathbb{C}^A(\varphi)$, $\mathbb{C}^R(\varphi)$, and $\mathbb{C}^I(\varphi)$, starting from each state i , where φ is either a bounded until formula, $a\text{U}^{[h,k]}b$, or an unbounded until formula, $a\text{U}^{[h,\infty)}b$. We stress that the lower time bound h may be positive. We first handle the case where h is 0, then explain how to propagate the computation backward when h is positive.

For path formulas φ of the form $a\mathbf{U}^{[0,k]}b$ or $a\mathbf{U}^{[0,\infty)}b$, let $B = \llbracket b \rrbracket$ and partition the state space $\mathcal{S}$ into “good” states $\mathcal{S}_g = \llbracket \mathbf{E}\varphi \rrbracket \supseteq B$, with at least one path satisfying φ , and “bad” states $\mathcal{S}_n = \mathcal{S} \setminus \mathcal{S}_g$. Here, $\mathbf{E}\varphi$ is the existential CTL state formula corresponding to path formula φ . The set $\mathcal{S}_g$ can be built using standard CTL algorithms [17], starting in B and stepping backward through predecessors satisfying a , at most k times for $a\mathbf{U}^{[0,k]}b$, until reaching a fixpoint for $a\mathbf{U}^{[0,\infty)}b$.

For $i \in \mathcal{S}_n$, $\mathbb{C}^A(\varphi)_i = \mathbb{C}^R(\varphi)_i = \mathbb{C}^I(\varphi)_i = \perp$, as no path from i satisfies φ . For states in $\mathcal{S}_g$, we build a modified DMRM $\mathcal{D}'$ from $\mathcal{D}$ by making all states in $\mathcal{S}_n \cup B$ absorbing. The remaining states $\mathcal{T}' = \mathcal{S}_g \setminus B$ are transient in $\mathcal{D}'$. Let $\mathbf{P}'$ be the transition probability matrix of $\mathcal{D}'$. For states in $\mathcal{T}'$, a path satisfies $a\mathbf{U}^{[0,k]}b$ iff it reaches B within k steps; it satisfies $a\mathbf{U}^{[0,\infty)}b$ iff it reaches B .

From the $\mathbb{C}$ operators semantics, if $i \in B$, $\mathbb{C}^A(\varphi)_i = 0$, $\mathbb{C}^R(\varphi)_i = \rho[i]$, $\mathbb{C}^I(\varphi)_i = 0$: the formula is already satisfied at time 0, thus no accumulated or impulse reward is accrued before satisfaction, while the reward rate upon satisfaction is $\rho[i]$.

Computing $\mathbb{C}^A$. The numerator in the formula for $\mathbb{C}^A$ in Fig. 1 is the expected accumulated reward over satisfying paths only, not over all paths. Thus, any reward collected before reaching B must be weighted by the probability that the remaining suffix of the path still satisfies the until formula.

For $m \in \mathbb{N}$, let $p_m[i]$ be the probability that, starting from state i , the path formula $a\mathbf{U}^{[0,m]}b$ is satisfied. Let $x_m[i]$ be the corresponding unnormalized numerator for $\mathbb{C}^A$, i.e., $x_m[i] = \sum_{\sigma \in \llbracket a\mathbf{U}^{[0,m]}b \rrbracket_i} \Pr(\sigma) \cdot \rho(\sigma)$. Thus, when $p_m[i] > 0$, $\mathbb{C}^A(a\mathbf{U}^{[0,m]}b)_i = x_m[i]/p_m[i]$. When $p_m[i] = 0$, instead, $\mathbb{C}^A(a\mathbf{U}^{[0,m]}b)_i = \perp$.

For all $m \geq 0$, $p_m[B] = \mathbf{1}$ and $x_m[B] = \mathbf{0}$ while $p_m[\mathcal{S}_n] = \mathbf{0}$ and $x_m[\mathcal{S}_n] = \mathbf{0}$. For states in $\mathcal{T}'$, we need to iterate. The base case is $p_0[\mathcal{T}'] = \mathbf{0}$ and $x_0[\mathcal{T}'] = \mathbf{0}$. Then, if $\varphi = a\mathbf{U}^{[0,k]}b$, for $1 \leq m \leq k$, the denominator satisfies

$$p_m[\mathcal{T}'] = \mathbf{P}'[\mathcal{T}', B] \cdot \mathbf{1} + \mathbf{P}'[\mathcal{T}', \mathcal{T}'] \cdot p_{m-1}[\mathcal{T}'] \quad (1)$$

and the unnormalized accumulated-reward numerator satisfies

$$\begin{aligned} x_m[\mathcal{T}'] = & \rho[\mathcal{T}'] \bullet p_m[\mathcal{T}'] + (\mathbf{P}'[\mathcal{T}', \mathcal{T}'] \bullet \rho[\mathcal{T}', \mathcal{T}']) \cdot p_{m-1}[\mathcal{T}'] \\ & + (\mathbf{P}'[\mathcal{T}', B] \bullet \rho[\mathcal{T}', B]) \cdot \mathbf{1} + \mathbf{P}'[\mathcal{T}', \mathcal{T}'] \cdot x_{m-1}[\mathcal{T}']. \end{aligned} \quad (2)$$

Thus, if $\varphi = a\mathbf{U}^{[0,k]}b$, we have $\mathbb{C}^A(\varphi)_i = x_k[i]/p_k[i]$ for $i \in \mathcal{S}_g$ with $p_k[i] > 0$, while $\mathbb{C}^A(\varphi)_i = \perp$ if $p_k[i] = 0$. Factor p_{m-1} in Eq. 2 is essential: a reward is collected at the current step only for paths whose suffix still satisfies φ .

If $\varphi = a\mathbf{U}^{[0,\infty)}b$, instead, the denominator is the least nonnegative solution of $p[\mathcal{T}'] = \mathbf{P}'[\mathcal{T}', B] \cdot \mathbf{1} + \mathbf{P}'[\mathcal{T}', \mathcal{T}'] \cdot p[\mathcal{T}']$, i.e., of $(I - \mathbf{P}'[\mathcal{T}', \mathcal{T}']) \cdot p[\mathcal{T}'] = \mathbf{P}'[\mathcal{T}', B] \cdot \mathbf{1}$, while the corresponding numerator is the least nonnegative solution of

$$(I - \mathbf{P}'[\mathcal{T}', \mathcal{T}']) \cdot x[\mathcal{T}'] = \rho[\mathcal{T}'] \bullet p[\mathcal{T}'] + (\mathbf{P}'[\mathcal{T}', \mathcal{T}'] \bullet \rho[\mathcal{T}', \mathcal{T}']) \cdot p[\mathcal{T}'] + (\mathbf{P}'[\mathcal{T}', B] \bullet \rho[\mathcal{T}', B]) \cdot \mathbf{1}.$$

Thus, $\mathbb{C}^A(\varphi)_i = x[i]/p[i]$ for $i \in \mathcal{S}_g$ with $p[i] > 0$, while $\mathbb{C}^A(\varphi)_i = \perp$ if $p[i] = 0$.

If the lower bound h is positive, we now show how to reduce the computation to the previous case, and then propagate the result backward h steps.

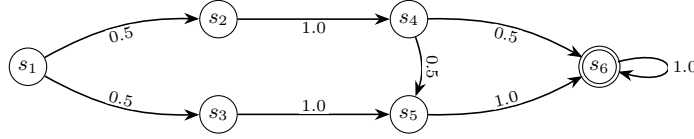


Fig. 2: Example DMRM on which we compute $\mathbb{C}^A(aU^{[2,4]}b)$.

Let φ_0 be $aU^{[0,k-h]}b$ if $\varphi = aU^{[h,k]}b$, and $aU^{[0,\infty)}b$ if $\varphi = aU^{[h,\infty)}b$, and compute the “suffix quantities” for φ_0 : the denominator vector $p'[\mathcal{S}]$ and the unnormalized accumulated-reward numerator vector $x'[\mathcal{S}]$, using the method for a 0 lower bound (as before, $p'[i] = 0$ and $x'[i] = 0$ for states i that cannot satisfy φ_0 , while $p'[i] = 1$ and $x'[i] = 0$ for $i \in \llbracket b \rrbracket$). Then, let $\varphi^{h \leftarrow 0}$ be φ with its lower time bound set to 0, i.e., $aU^{[0,k]}b$ or $aU^{[0,\infty)}b$, and build a second modified DMRM $\mathcal{D}''$, also derived from $\mathcal{D}$ with state space partitioned into $\mathcal{S}_g'' = \llbracket E \varphi^{h \leftarrow 0} \rrbracket$ and $\mathcal{S}_n'' = \mathcal{S} \setminus \mathcal{S}_g''$. States $\mathcal{T}'' = \llbracket a \rrbracket \cap \mathcal{S}_g''$ are transient in $\mathcal{D}''$, while all other states are made absorbing. Thus, during the first h “prefix steps”, the path is allowed to visit only states satisfying a . To enforce this prefix of length h , let $p_h''[\mathcal{S}] = p'[\mathcal{S}]$ and $x_h''[\mathcal{S}] = x'[\mathcal{S}]$. Then, for $\ell = h, h-1, \dots, 1$, compute $p_{\ell-1}''[\mathcal{T}''] = \mathbf{P}''[\mathcal{T}'', \mathcal{S}] \cdot p_\ell''[\mathcal{S}]$ and $x_{\ell-1}''[\mathcal{T}''] = \rho[\mathcal{T}''] \bullet p_{\ell-1}''[\mathcal{T}''] + (\mathbf{P}''[\mathcal{T}'', \mathcal{S}] \bullet \rho[\mathcal{T}'', \mathcal{S}]) \cdot p_\ell''[\mathcal{S}] + \mathbf{P}''[\mathcal{T}'', \mathcal{S}] \cdot x_\ell''[\mathcal{S}]$. If $i \notin \mathcal{T}''$, let $p_{\ell-1}''[i] = 0$ and $x_{\ell-1}''[i] = 0$ during the prefix iterations. Finally, $\mathbb{C}^A(\varphi)_i = x_0''[i]/p_0''[i]$ if $p_0''[i] > 0$, else $\mathbb{C}^A(\varphi)_i = \perp$.

As an example, we show how to compute $\mathbb{C}^A(aU^{[2,4]}b)$ on DMRM $\mathcal{D}$ in Fig. 2, where states s_1, s_2, s_3, s_4, s_5 satisfy a and state s_6 satisfies b . The reward vector is $\rho[s]=1$ for $s \in \{s_1, s_2, s_3, s_4, s_5\}$ and $\rho[s_6]=0$, while all impulse rewards are zero. For all vectors below, we use the state order $(s_1, s_2, s_3, s_4, s_5)$.

First, we compute the suffix quantities for $\varphi_0 = aU^{[0,2]}b$. The satisfaction-probability vectors are $p_0 = [0,0,0,0,0]$, $p_1 = [0,0,0,0.5,1]$, and $p_2 = [0,0.5,1,1,1]$, the corresponding unnormalized accumulated-reward numerator vectors are $x_0 = [0,0,0,0,0]$, $x_1 = [0,0,0,0.5,1]$, and $x_2 = [0,1,2,1.5,1]$. For example, from s_2 , the suffix formula $aU^{[0,2]}b$ is satisfied only on path $s_2 \rightarrow s_4 \rightarrow s_6$, which has probability 0.5 and accumulated reward 2. Hence $p_2[s_2] = 0.5$ and $x_2[s_2] = 0.5 \cdot 2 = 1$.

To obtain $\mathbb{C}^A(aU^{[2,4]}b)$, we propagate these suffix vectors backward through the prefix of length $h=2$ (for full vectors, the suffix initialization includes $p_2''[s_6]=1$, and $x_2''[s_6]=0$, since a suffix starting in a b -state is satisfied immediately and collects no additional accumulated reward). Thus, $p_2''=p_2$, and $x_2''=x_2$ for $s_1, \dots, s_5$.

The first prefix step gives $p_1'' = [0.75, 1, 1, 1, 1]$, and $x_1'' = [2.25, 2.5, 2, 1.5, 1]$, the second prefix step gives $p_0'' = [1, 1, 1, 0.5, 0]$, and $x_0'' = [3.25, 2.5, 2, 1, 0]$. Then, the conditional accumulated rewards are obtained by dividing the unnormalized numerator by the satisfaction probability: $\mathbb{C}^A(\varphi)_i = x_0''[i]/p_0''[i]$ whenever $p_0''[i] > 0$. Thus, $\mathbb{C}^A(aU^{[2,4]}b) = [3.25, 2.5, 2, 2, \perp]$ for states $(s_1, s_2, s_3, s_4, s_5)$, respectively. Note that the value for s_4 is 2, not 1.5. Transition $s_4 \rightarrow s_6$ reaches b at time 1, before the lower bound 2, and since s_6 does not satisfy a , this branch cannot satisfy $aU^{[2,4]}b$. The only satisfying path from s_4 is $s_4 \rightarrow s_5 \rightarrow s_6$, which has probability 0.5 and accumulated reward 2. Hence $x_0''[s_4] = 0.5 \cdot 2 = 1$ and $p_0''[s_4] = 0.5$, thus $\mathbb{C}^A(\varphi)_{s_4} = 1/0.5 = 2$. State s_5 has no satisfying path for $aU^{[2,4]}b$, so $\mathbb{C}^A(\varphi)_{s_5} = \perp$.

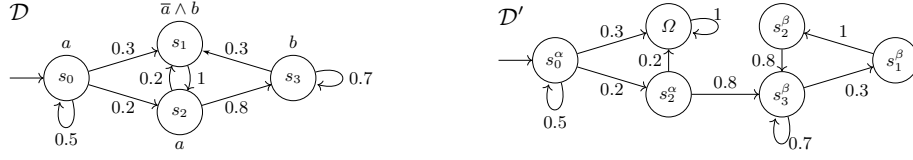


Fig. 3: DMRM $\mathcal{D}$ and derived DMRM $\mathcal{D}'$, for $\varphi = aU^{[h,k]}b$ or $\varphi = aU^{[h,\infty)}b$.

Computing $\mathbb{C}^R$ and $\mathbb{C}^I$. The denominator for $\mathbb{C}^R$ and $\mathbb{C}^I$ is as for $\mathbb{C}^A$.

For $\mathbb{C}^R$, the numerator records the reward rate of the first b -state satisfying φ . Then, $x_0^R[\mathcal{T}'] = \mathbf{0}$ and $x_m^R[\mathcal{T}'] = \mathbf{P}'[\mathcal{T}', B] \cdot \rho[B] + \mathbf{P}'[\mathcal{T}', \mathcal{T}'] \cdot x_{m-1}^R[\mathcal{T}']$, for $1 \leq m \leq k$. Thus, when $\varphi = aU^{[0,k]}b$, $\mathbb{C}^R(\varphi)_i = x_k^R[i]/p_k[i]$ for $i \in \mathcal{S}_g$ with $p_k[i] > 0$.

For $\mathbb{C}^I$, the numerator records the impulse reward of the first transition entering a b -state. Then, $x_0^I[\mathcal{T}'] = \mathbf{0}$ and $x_m^I[\mathcal{T}'] = (\mathbf{P}'[\mathcal{T}', B] \bullet \rho[\mathcal{T}', B]) \cdot \mathbf{1} + \mathbf{P}'[\mathcal{T}', \mathcal{T}'] \cdot x_{m-1}^I[\mathcal{T}']$, for $1 \leq m \leq k$. Thus, when $\varphi = aU^{[0,k]}b$, $\mathbb{C}^I(\varphi)_i = x_k^I[i]/p_k[i]$ for $i \in \mathcal{S}_g$ with $p_k[i] > 0$.

When $\varphi = aU^{[0,\infty)}b$, the numerators for $\mathbb{C}^R$ and $\mathbb{C}^I$ are, respectively, the least nonnegative solutions of $(I - \mathbf{P}'[\mathcal{T}', \mathcal{T}']) \cdot x^R[\mathcal{T}'] = \mathbf{P}'[\mathcal{T}', B] \cdot \rho[B]$ and of $(I - \mathbf{P}'[\mathcal{T}', \mathcal{T}']) \cdot x^I[\mathcal{T}'] = (\mathbf{P}'[\mathcal{T}', B] \bullet \rho[\mathcal{T}', B]) \cdot \mathbf{1}$. Then, $\mathbb{C}^R(\varphi)_i = x^R[i]/p[i]$, and $\mathbb{C}^I(\varphi)_i = x^I[i]/p[i]$, for $i \in \mathcal{S}_g$ with $p[i] > 0$; while they are $\perp$ if $p[i] = 0$.

The computation for the case $h > 0$ is exactly analogous as the one for $\mathbb{C}^A$.

4.3 Computing $\mathbb{T}^A$, $\mathbb{T}^R$, and $\mathbb{T}^I$

Formulas $\mathbb{T}^A$, $\mathbb{T}^R$, and $\mathbb{T}^I$ observe the behavior of $\mathcal{D}$ up to, or at, time t , conditioned on the path satisfying $\varphi = aU^{[h,k]}b$ or $\varphi = aU^{[h,\infty)}b$. We build expanded DMRM $\mathcal{D}'$ to keep track of the satisfaction status of φ along paths: (i) an “ α ” state represents executions that have not yet satisfied φ , but can still satisfy it; (ii) a “ β ” state represents executions that have already satisfied φ ; (iii) executions that can no longer satisfy φ are merged into absorbing macro-state Ω .

Consider first the case $h = 0$. $\mathcal{D}'$ has state space $\mathcal{S}' \subseteq (\mathcal{S} \times \{\alpha, \beta\}) \cup \{\Omega\}$. Let $B = \llbracket b \rrbracket$. For each $i \in \mathcal{S}$, add i^α to $\mathcal{S}'$ iff $i \models a \wedge \neg b \wedge E\varphi$ (the α -copy represents executions that have not yet satisfied φ , but can still do so while preserving the required a -prefix) and add i^β to $\mathcal{S}'$ iff i is reachable from some state in B , including the case $i \in B$ itself (the β -copy represents executions that have already satisfied φ). Finally, Ω is an absorbing macro-state representing executions that can no longer satisfy φ . A state i may have both an α -copy and a β -copy. The α -copy is used when i is reached before φ has been satisfied, whereas the β -copy is used when i is reached after φ has already been satisfied. Thus, $|\mathcal{S}'| \leq 2|\mathcal{S}| + 1$, up to two copies of each original state, plus state Ω . Fig. 3 shows a small example.

The transition probability matrix $\mathbf{P}'$ of $\mathcal{D}'$ can be partitioned into blocks, with entries related to those of the original $\mathbf{P}$ as follows:

$$\mathbf{P}' = \begin{matrix} & \alpha & \beta & \Omega \\ \begin{matrix} \alpha \\ \beta \\ \Omega \end{matrix} & \begin{bmatrix} \mathbf{P}_{\alpha,\alpha} & \mathbf{P}_{\alpha,\beta} & \mathbf{P}_{\alpha,\Omega} \\ \mathbf{0} & \mathbf{P}_{\beta,\beta} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 1 \end{bmatrix} \end{matrix},$$

since, once φ is satisfied, $\mathcal{D}'$ remains in β -states, $\forall i^\beta \in \mathcal{S}', \forall j^\beta \in \mathcal{S}', \mathbf{P}'[i^\beta, j^\beta] = \mathbf{P}[i, j]$, while macro-state Ω is absorbing, $\mathbf{P}'[\Omega, \Omega] = 1$. For transitions leaving an α -state, the successor copy is instead determined by whether the current transition satisfies φ . If the destination satisfies b , then the transition satisfies φ and enters a β -state: $\forall i^\alpha \in \mathcal{S}', \forall j \in B$ with $j^\beta \in \mathcal{S}', \mathbf{P}'[i^\alpha, j^\beta] = \mathbf{P}[i, j]$. If the successor does not satisfy b but the path can still satisfy φ , then the execution remains in α -states: $\forall i^\alpha \in \mathcal{S}', \forall j \notin B$ with $j^\alpha \in \mathcal{S}', \mathbf{P}'[i^\alpha, j^\alpha] = \mathbf{P}[i, j]$. All remaining transitions from an α -state go to Ω : $\forall i^\alpha \in \mathcal{S}', \mathbf{P}'[i^\alpha, \Omega] = \sum_{j \in S \setminus B, j^\alpha \notin \mathcal{S}'} \mathbf{P}[i, j]$. All other entries not specified above are zero.

Thus, a state may have both an α -copy and a β -copy, but a transition from an α -state chooses exactly one successor: the β -copy of the successor in original $\mathcal{D}$ if the current transition reaches a b -state and therefore satisfies the until formula; the α -copy of the successor if it can still satisfy the formula; otherwise, it goes to Ω . Consequently, every probability $\mathbf{P}[i, j]$ leaving an α -state is assigned exactly once, either to j^α , to j^β , or to Ω , so every row of $\mathbf{P}'$ sums to one. The reward structure of $\mathcal{D}$ lifts to $\mathcal{D}'$ in the natural way: each copy of state i inherits $\rho[i]$ as its rate reward, and each transition in $\mathcal{D}'$ inherits the impulse reward of the corresponding transition in $\mathcal{D}$. The macro-state Ω has zero rate reward, and transitions into Ω carry zero impulse reward.

Computing $\mathbb{T}^R$ and $\mathbb{T}^I$. Consider observation time $t < \infty$ and $\varphi = a\mathbf{U}^{[0, k]}b$ or $\varphi = a\mathbf{U}^{[0, \infty)}b$. For a finite upper bound k , let $\mathbf{b}_m[\mathcal{S}']$ denote the probability that, starting from each state of the expanded DMRM $\mathcal{D}'$, the path reaches the β -states in the m remaining steps. Thus, for $m \leq 0$, $\mathbf{b}_m[i^\alpha] = 0$, $\mathbf{b}_m[i^\beta] = 1$, $\mathbf{b}_m[\Omega] = 0$, and, for $m > 0$, $\mathbf{b}_m = \mathbf{P}' \cdot \mathbf{b}_{m-1}$. For unbounded until, let $\mathbf{b}_\infty = \lim_{m \rightarrow \infty} \mathbf{b}_m$.

For finite k , vector $\mathbf{b}_{k-t}$ gives, from the state at time t , the probability that the path has already satisfied φ or will satisfy it by the upper bound k . If $t > k$, then, this gives $\mathbf{b}_{k-t}[i^\alpha] = 0$ and $\mathbf{b}_{k-t}[i^\beta] = 1$, as desired.

For $\mathbb{T}^R(\varphi, t)$ on $\mathcal{D}'$, the unnormalized numerator $\mathbf{x}_0^R$ is $(\mathbf{P}')^t \cdot (\rho[\mathcal{S}'] \bullet \mathbf{b}_{k-t})$ for finite k , and $(\mathbf{P}')^t \cdot (\rho[\mathcal{S}'] \bullet \mathbf{b}_\infty)$ for unbounded until. For $\mathbb{T}^I$, the impulse reward is 0 if $t = 0$; for $t \geq 1$, the unnormalized numerator $\mathbf{x}_0^I$ is $(\mathbf{P}')^{t-1} \cdot ((\mathbf{P}' \bullet \rho[\mathcal{S}', \mathcal{S}'] \cdot \mathbf{b}_{k-t})$ for finite k , and $(\mathbf{P}')^{t-1} \cdot ((\mathbf{P}' \bullet \rho[\mathcal{S}', \mathcal{S}'] \cdot \mathbf{b}_\infty)$ for unbounded until.

We can also compute these vectors by backward iteration. For $\mathbb{T}^R$, initialize $\mathbf{x}_t = \rho[\mathcal{S}'] \bullet \mathbf{b}_{k-t}$, or $\mathbf{x}_t = \rho[\mathcal{S}'] \bullet \mathbf{b}_\infty$ for unbounded until, and iterate $\mathbf{x}_{n-1} = \mathbf{P}' \cdot \mathbf{x}_n$, for $n = t, t-1, \dots, 1$. For $\mathbb{T}^I$ with $t \geq 1$, initialize $\mathbf{x}_t = (\mathbf{P}' \bullet \rho[\mathcal{S}', \mathcal{S}'] \cdot \mathbf{b}_{k-t})$, or $\mathbf{x}_t = (\mathbf{P}' \bullet \rho[\mathcal{S}', \mathcal{S}'] \cdot \mathbf{b}_\infty)$ for unbounded until, and again iterate $\mathbf{x}_{n-1} = \mathbf{P}' \cdot \mathbf{x}_n$, for $n = t-1, t-2, \dots, 1$. The resulting vector $\mathbf{x}_0$ is an unnormalized numerator.

We now must map the numerator vector back to the original DMRM $\mathcal{D}$ and divide by the probability of satisfying φ . If we let

$$\hat{x}[i] = \mathbf{x}_0[i^\alpha] \text{ if } i^\alpha \in \mathcal{S}', \quad \hat{x}[i] = \mathbf{x}_0[i^\beta] \text{ if } i^\alpha \notin \mathcal{S}' \wedge i^\beta \in \mathcal{S}', \quad \hat{x}[i] = \perp \text{ if } i^\alpha \notin \mathcal{S}' \wedge i^\beta \notin \mathcal{S}',$$

then, $\mathbf{y}[i] = \hat{x}[i] / \mathbb{P}(\varphi)_i$ if $\mathbb{P}(\varphi)_i > 0$, and $\mathbf{y}[i] = \perp$ if $\mathbb{P}(\varphi)_i = 0$, gives the correct value of either $\mathbb{T}^R(\varphi, t)$ or $\mathbb{T}^I(\varphi, t)$, depending on which numerator vector is used.

If $t = \infty$, we compute the limiting reward in the β -states of $\mathcal{D}'$, where a path that satisfies φ eventually enters the β -states, while paths that cannot satisfy

φ enter Ω . Consider then the recurrent classes of β -states, letting $\mathbf{b}^{\text{rec}}[S']$ be the indicator vector of states in β -recurrent classes, i.e., $\mathbf{b}^{\text{rec}}[i^\beta] = 1$ if i^β is in a recurrent class of $\mathcal{D}'$, and $\mathbf{b}^{\text{rec}}[j] = 0$ otherwise. The unnormalized numerator for $\mathbb{T}^{\text{R}}(\varphi, \infty)$ is $\mathbf{x}_0^{\text{R}} = (\mathbf{P}')^\infty \cdot (\rho[S'] \bullet \mathbf{b}^{\text{rec}})$, where $(\mathbf{P}')^\infty$ is the limiting transition matrix, using the Cesàro limit if $\mathcal{D}'$ contains periodic recurrent classes.

For $\mathbb{T}^{\text{I}}(\varphi, \infty)$, the unnormalized numerator is $\mathbf{x}_0^{\text{I}} = (\mathbf{P}')^\infty \cdot ((\mathbf{P}' \bullet \rho[S', S']) \cdot \mathbf{b}^{\text{rec}})$. As in the finite-time case, the resulting numerator vector is mapped back to the original state space and divided by $\mathbb{P}(\varphi)_i$ whenever $\mathbb{P}(\varphi)_i > 0$.

Once we have determined vector $\mathbf{x}_0$ on DMRM $\mathcal{D}'$, we obtain the desired result, vector $\mathbf{y}$ on the original DMRM $\mathcal{D}$, as follows: for each $i \in \mathcal{S}$, if $i^\alpha \in S'$, let $\mathbf{y}[i] = \mathbf{x}_0[i^\alpha]$; if $i^\alpha \notin S'$ and $i^\beta \in S'$, let $\mathbf{y}[i] = \mathbf{x}_0[i^\beta]$; if $i^\alpha \notin S'$ and $i^\beta \notin S'$, let $\mathbf{y}[i] = \perp$, as there are no paths from i satisfying φ .

For path formulas $\varphi = a\mathbf{U}^{[h,k]}b$ or $\varphi = a\mathbf{U}^{[h,\infty)}b$ with $h > 0$, we separate the prefix of length h from the suffix beginning at time h . Let $\varphi_0 = a\mathbf{U}^{[0,k-h]}b$ if $\varphi = a\mathbf{U}^{[h,k]}b$ and $\varphi_0 = a\mathbf{U}^{[0,\infty)}b$ if $\varphi = a\mathbf{U}^{[h,\infty)}b$, build the expanded DMRM $\mathcal{D}'$ for φ_0 , and compute the corresponding suffix satisfaction-probability vector. Then, build a second modified DMRM $\mathcal{D}''$ over the original state space $\mathcal{S}$, where states not satisfying a are absorbing. This DMRM is used to propagate information backward through the a -prefix. First compute the prefix satisfaction-probability vectors: let $p_h''[\mathcal{S}]$ be the probability of satisfying the suffix formula φ_0 at time h from each state. Then, for $\ell = h, h-1, \dots, 1$, compute $p_{\ell-1}''[\mathcal{S}] = \mathbf{P}'' \cdot p_\ell''[\mathcal{S}]$, setting entries of states that violate the required a -prefix to 0.

If $t \geq h$, the reward observation occurs in the suffix part. First, compute the suffix numerator vector for time $t-h$ using the $h = 0$ construction above. Let this vector be $\mathbf{x}'_h$ on the expanded suffix DMRM $\mathcal{D}'$, then map $\mathbf{x}'_h$ back to a vector over the original state space at time h , and propagate it backward through the prefix: $\mathbf{x}_{h-1}''[i] = \sum_{j \in \mathcal{S}} \mathbf{P}''[i, j] \cdot \mathbf{x}'_h[j]$, for all a -states i , then let $\mathbf{x}_{\ell-1}'' = \mathbf{P}'' \cdot \mathbf{x}_\ell''$ for $\ell = h-1, h-2, \dots, 1$. The final numerator is $\mathbf{x}_0''$.

If $t < h$, the reward observation occurs inside the prefix. For $\mathbb{T}^{\text{R}}$, compute the numerator by placing the reward at prefix time t : $\mathbf{x}_t'' = \rho[\mathcal{S}] \bullet p_t''[\mathcal{S}]$, and propagating it backward: $\mathbf{x}_{\ell-1}'' = \mathbf{P}'' \cdot \mathbf{x}_\ell''$, for $\ell = t, t-1, \dots, 1$. For $\mathbb{T}^{\text{I}}$, if $t = 0$, the numerator is the zero vector; if $1 \leq t < h$, the numerator is computed using the impulse reward on the transition into time t : $\mathbf{x}_t'' = (\mathbf{P}'' \bullet \rho[\mathcal{S}, \mathcal{S}]) \cdot p_t''[\mathcal{S}]$, followed by the same backward propagation to time 0: $\mathbf{x}_{\ell-1}'' = \mathbf{P}'' \cdot \mathbf{x}_\ell''$ for $\ell = t-1, t-2, \dots, 1$.

In all cases, the final conditional value is the resulting unnormalized numerator divided by the satisfaction probability: $\mathbb{T}^{\text{R}}(\varphi, t)_i = \mathbf{x}_0''[i]/p_0''[i]$ and $\mathbb{T}^{\text{I}}(\varphi, t)_i = \mathbf{x}_0''[i]/p_0''[i]$, if $p_0''[i] > 0$; if $p_0''[i] = 0$, the value is $\perp$.

Computing $\mathbb{T}^{\text{A}}$. Consider first a finite observation time $t < \infty$. By Def. 8, the accumulated reward up to time t is the sum of the reward rates at times $0, \dots, t-1$ and the impulse rewards on the transitions at times $1, \dots, t$. Thus, $\mathbb{T}^{\text{A}}(\varphi, t)_{i_0} = \sum_{n=0}^{t-1} \mathbb{T}^{\text{R}}(\varphi, n)_{i_0} + \sum_{n=1}^t \mathbb{T}^{\text{I}}(\varphi, n)_{i_0}$, but there is no need to separately compute $\mathbb{T}^{\text{R}}(\varphi, n)$ and $\mathbb{T}^{\text{I}}(\varphi, n)$: we can simply compute their unnormalized numerator contributions on the expanded DMRM $\mathcal{D}'$ and divide by $\mathbb{P}(\varphi)_i$ at the end.

For the unnormalized accumulated rate-reward numerator, let $\mathbf{y}_t^{\text{R}} = \mathbf{0}$ and iterate backward: $\mathbf{y}_n^{\text{R}} = (\rho[S'] \bullet \mathbf{b}_{k-n}) + \mathbf{P}' \cdot \mathbf{y}_{n+1}^{\text{R}}$, for $n = t-1, t-2, \dots, 0$. Here, $\mathbf{b}_{k-n}$

is the vector defined above: its entry at a state of $\mathcal{D}'$ is the probability that the path has already satisfied φ or will satisfy it by the upper bound k , starting from that state at time n . For an unbounded until, $\mathbf{b}_{k-n}$ is replaced by $\mathbf{b}_\infty$.

Similarly, to compute the unnormalized accumulated impulse-reward numerator, let $\mathbf{y}_{t+1}^I = \mathbf{0}$ and iterate backward: $\mathbf{y}_n^I = (\mathbf{P}' \bullet \rho[\mathcal{S}', \mathcal{S}']) \cdot \mathbf{b}_{k-n} + \mathbf{P}' \cdot \mathbf{y}_{n+1}^I$, for $n=t, t-1, \dots, 1$, where the first term is a vector whose entry for state $i' \in \mathcal{S}'$ is $\sum_{j' \in \mathcal{S}'} \mathbf{P}'[i', j'] \rho[i', j'] \mathbf{b}_{k-n}[j']$. Thus, the impulse reward earned on the transition from time $n-1$ to time n is counted only if the suffix from destination state j' can still satisfy φ . For an unbounded until, $\mathbf{b}_{k-n}$ is again replaced by $\mathbf{b}_\infty$.

The unnormalized numerator for $\mathbb{T}^A(\varphi, t)$ on $\mathcal{D}'$ is then $\mathbf{y}^A = \mathbf{y}_0^R + \mathbf{y}_1^I$. After mapping $\mathbf{y}^A$ back to the original state space $\mathcal{S}$ as described above, we divide by the satisfaction probability. That is, for each original state i with $\mathbb{P}(\varphi)_i > 0$, $\mathbb{T}^A(\varphi, t)_i = \hat{y}^A[i] / \mathbb{P}(\varphi)_i$, where $\hat{y}^A[i]$ is the value mapped back from the corresponding α - or β -copy of i in $\mathcal{D}'$. If $\mathbb{P}(\varphi)_i = 0$, then $\mathbb{T}^A(\varphi, t)_i = \perp$.

Finally, consider the computation of $\mathbb{T}^A(\varphi, \infty)$. In the expanded DMRM $\mathcal{D}'$, any path either eventually reaches Ω , which represents paths that do not satisfy φ , or eventually remains in β -states, which represents paths that have satisfied φ . Thus, for paths satisfying φ , the long-run behavior is determined by the recurrent classes contained in the β -states.

If a β -recurrent class contains a state with nonzero reward rate or a transition with nonzero impulse reward, then every path that reaches this class accumulates infinite reward. Consequently, any state from which such a class is reached with positive probability has $\mathbb{T}^A(\varphi, \infty)_i = \infty$ after conditioning on φ .

For the remaining states, all reachable β -recurrent classes have zero reward rates and zero impulse rewards. We merge these zero-reward β -recurrent classes into a single absorbing “good” macro-state Γ . States that can reach a positive-reward β -recurrent class have already been classified as having value ∞ . Let $\mathcal{D}'_\infty$ denote the resulting modified DMRM. Then, the finite value of $\mathbb{T}^A(\varphi, \infty)$ for the remaining states is obtained by computing $\mathbb{C}^A(\text{trueUI}\Gamma)$ on $\mathcal{D}'_\infty$, with the reward structure inherited from $\mathcal{D}'$. Γ represents satisfying paths eventually entering a zero-reward recurrent class, while Ω represents paths not satisfying φ .

4.4 Complexity

We briefly consider the time complexity of computing our new operators. Let $|\mathcal{S}|$ be the number of states, $|\mathbf{P}|$ be the number of non-zero entries in the transition probability matrix, and h and k be the lower and upper time bounds of the until formula φ . According to [17], standard CTL backward reachability takes $O(k|\mathbf{P}|)$ for bounded φ and $O(|\mathbf{P}|)$ for unbounded φ .

To compute the $\mathbb{C}^A$ operator with a bounded $\varphi = a\mathbf{U}^{[h,k]}b$, each iteration of the recurrence for p_m and x_m in $\mathcal{D}'$ performs a matrix-vector multiplication of size $|\mathcal{T}'|$, with complexity $O(|\mathbf{P}|)$, plus $O(|\mathcal{T}'|)$ element-wise operations. If $h > 0$, the recurrence occurs $k-h$ times in $\mathcal{D}'$ and h times in $\mathcal{D}''$ for a total of $O(k|\mathbf{P}|)$ time. The $\mathbb{C}^R$ and $\mathbb{C}^I$ operators use the same p_m vectors computation as $\mathbb{C}^A$ and one additional backward iteration each, thus have the same cost as $\mathbb{C}^A$.

For an unbounded until formula $\varphi = a\mathbf{U}^{[h,\infty)}b$, if N iterations are required to converge, the complexity is $O(N|\mathbf{P}|)$. If $h > 0$, the additional cost is $O(h|\mathbf{P}|)$ in $\mathcal{D}''$. For the $\mathbb{T}^A$, $\mathbb{T}^R$, and $\mathbb{T}^I$ operators, building $\mathcal{D}'$ with α and β copies of the states and macro-state Ω takes $O(|\mathbf{P}|)$ time and $O(|\mathcal{S}|)$ space because $|\mathcal{S}'| \leq 2|\mathcal{S}| + 1$ and each state contributes at most one α -state and one β -state. For bounded φ and finite time t , computing b_m for $m = 0, \dots, k - t$ and the backward iterations for y_n^R or y_n^I each perform one matrix-vector product on $\mathcal{D}'$ per step, for a total of $O(\max(k, t)|\mathbf{P}|)$ time; if $h > 0$, there is an additional cost $O(h|\mathbf{P}|)$ for the prefix iteration on $\mathcal{D}''$. For unbounded φ , the same backward iteration uses the limiting vector b_∞ obtained from solving a linear system of size $|\mathcal{S}'|$, with cost $O(N|\mathbf{P}|)$. For $\mathbb{T}^A(\varphi, \infty)$, identifying the recurrent classes of $\mathcal{D}'$ via Tarjan's SCC algorithm [31] takes $O(|\mathbf{P}|)$, after which the computation reduces to a single $\mathbb{C}^A(\text{true} \mathbf{U} F)$ solution on $\mathcal{D}'_\infty$, with the same asymptotic cost.

In other words, our new operators have computational cost analogous to that of the standard PCTL and PRCTL operators.

5 Two applications of our PRCTL extensions

5.1 Asynchronous leader election protocol

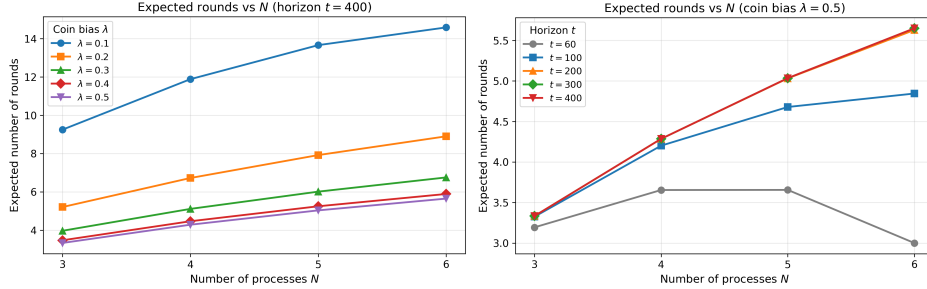
We study the expected number of rounds to elect a leader, $\mathbb{C}^A(\text{true} \mathbf{U} \text{elected})$, using the protocol of [22] to elect a (unique) leader by asynchronously sending messages around a ring of N processes. All processes are initially active and each active process n performs the following cycle of steps until either it becomes inactive (thus it only relays messages) or it becomes the leader.

1. Process n flips a coin and sends its choice to the next process along the ring.
2. If process n chose 0 and it receives a 1, it becomes inactive,
3. otherwise, it sends a probe around the ring to check for other active processes:
 - (a) if there is none, process n becomes the leader,
 - (b) otherwise, process n (as all other active processes) repeats the cycle.

We vary the number N of processes in the ring from 3 to 6 and the probability λ for an active process to flip a 1, i.e., to run for election in any given round, from 0.1 to 0.9, and compute the expected number of rounds to elect a leader (this differs from the expected time to elect a leader because each round requires a number of steps that depends on the number of active processes). The PRISM [28] website models this protocol as a Markov decision process, we instead employ a DMRM. PRISM case studies use an impulse reward when a new round begins (equals the number of times a process receives a counter); we instead use a state reward and compute the equivalent expected reward rate for each state i as $\rho[i] = \sum_j \mathbf{P}[i, j] \cdot \rho[j]$. From Table 1, we see that, unsurprisingly, the expected number of rounds to elect a leader increases with the number N of processes; more interestingly, we also see that the least expected number of rounds to elect a leader is always achieved for $\lambda = 0.5$, regardless of N (indeed, it is symmetric around $\lambda = 0.5$, i.e., it has the same value for λ and $1 - \lambda$). The last column

Table 1: Experimental results for the leader election protocol model.

Model			Expected number of rounds when λ is									Time
N	States	Transitions	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	(sec.)
3	364	654	9.25	5.20	3.96	3.47	3.33	3.47	3.96	5.20	9.25	0.0034
4	3,172	7,144	11.99	6.72	5.11	4.46	4.28	4.46	5.11	6.72	11.99	0.0176
5	27,299	74,365	14.13	7.92	6.01	5.24	5.03	5.24	6.01	7.92	14.13	0.2403
6	237,656	760,878	15.83	8.91	6.75	5.89	5.64	5.89	6.75	8.91	15.96	2.5054


 Fig. 4: Expected number of rounds vs. number of processes for (left) $t = 400$ and $\lambda \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$; (right) $\lambda = 0.5$ and $t \in \{60, 100, 200, 300, 400\}$.

shows the time to compute the results when $\lambda = 0.5$. In addition to computing the expected number of rounds to elect a leader over an infinite horizon, we also consider a finite time horizon t . Fig. 4 shows the expected number of rounds as a function of N and λ (for $t = 400$) or of N and t (for $\lambda = 0.5$).

5.2 Automated Guidance Vehicle collision avoidance system

Our next application is inspired by an UPPAAL case study originally proposed for job scheduling and optimal path planning [13]. Consider a fleet of autonomous guided vehicles (AGVs) operating in a warehouse to move a workload of m items between locations. Each AGV x can be in one of five local modes: mv_x indicating moving, wt_x indicating waiting due to congestion, $cdti_x$ indicating collision due to an impact with another AGV, $dlvr_x$, indicating that one item has just been delivered, or dn_x , indicating that its entire workload has been completed.

Fig. 5a shows a DMRM modeling AGV x when $m = 1$, i.e., when it has only one item to deliver. If the AGV is in state mv_x , it transitions to wt_x with probability q , to $cdti_x$ with probability r , to $dlvr_x$ with probability p , or back to mv_x with the complementary probability $1 - p - q - r$. If the AGV is in state $cdti_x$, it returns to mv_x with probability 1. From state wt_x , it transitions to $cdti_x$ with probability s , or returns to mv_x with probability $1 - s$. In the experiments, we use $p = 0.6$, $q = 0.1$, $r = 0.05$, and $s = 0.02$.

When an AGV delivers an item (enters state $dlvr_x$), it earns a unit impulse reward. If additional items remain, the AGV returns to mv_x and continues the process; otherwise, it transitions to dn_x , where it remains with probability 1. Thus, in Fig. 5a (case $m = 1$), the only possible transition from $dlvr_x$ is to dn_x .

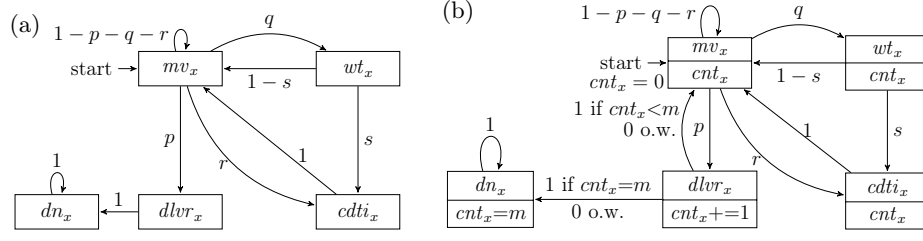


Fig. 5: AGV DMRM models: (a) base model for $m=1$. Upon entering $dlvr_x$, the AGV earns a unit impulse reward, then it transitions to dn_x , and remains there; (b) counter-augmented model for a general workload of m items.

If $m > 1$, the DMRM of AGV x also maintains a counter variable cnt_x , initially 0 and incremented by one whenever the AGV delivers an item. The maximum value of cnt_x is m , its total workload. To obtain a finite DMRM, the counter is unrolled: an AGV with workload m has $4m + 1$ local states, corresponding to the modes mv , wt , $cdti$, and $dlvr$ for counter values $0, \dots, m - 1$, plus the absorbing state dn with $cnt_x = m$. Fig. 5b shows this counter-augmented DMRM.

We then build a fleet model with n AGVs, each with workload m . In the full fleet DMRM, all AGVs take one local transition synchronously at each discrete time step. Let κ be the number of AGVs whose local mode is mv in the current global state. We use a traffic-scaled variant in which congestion and collision probabilities increase with the number of moving AGVs. Specifically,

$$q_{\text{eff}} = q \cdot \frac{\max(\kappa - 1, 0)}{\max(n - 1, 1)}, \quad r_{\text{eff}} = r \cdot \frac{\max(\kappa - 1, 0)}{\max(n - 1, 1)}, \quad s_{\text{eff}} = s \cdot \frac{\max(\kappa - 1, 0)}{\max(n - 1, 1)}.$$

Thus, when all AGVs are moving, $q_{\text{eff}} = q$, $r_{\text{eff}} = r$, and $s_{\text{eff}} = s$. The probability of remaining in mv is $1 - p - q_{\text{eff}} - r_{\text{eff}}$. Since $q_{\text{eff}} \leq q$ and $r_{\text{eff}} \leq r$, the base condition $p + q + r \leq 1$ ensures that this is a valid probability. For $n = 1$, $q_{\text{eff}} = r_{\text{eff}} = s_{\text{eff}} = 0$, matching the fact that a single AGV has no traffic interactions.

Integrating the counter into the model enables more complex queries, such as the expected reward conditioned on reaching states satisfying a global progress condition. To this end, we can define atomic proposition *success* to hold in global states where at least two thirds of the total workload has been delivered: $\sum_{1 \leq x \leq n} cnt_x \geq 2mn/3$, and atomic proposition *collide* to hold in global states where at least one AGV is in a collided state: $collide = \bigvee_{1 \leq x \leq n} cdti_x$.

We now explore performance-related queries concerning the expected number of delivered items. In all cases, all reward rates are 0 and the impulse reward of a global transition is the number of AGVs entering their respective local state $dlvr_x$ on that transition. Since each AGV remains in $dlvr_x$ for exactly one time step after each delivery, we could have equivalently used a reward structure where all impulse rewards are 0 and the reward rate of a global state is the number of AGVs currently in their local $dlvr_x$ state. We focus on the following queries:

- The expected accumulated reward up to time T over all paths:
 $\mathbb{C}^A(\varphi_1) = \mathbb{C}^A(\mathbf{true} \cup^{[T, T]} \mathbf{true}).$

Table 2: Results for the AGV case study.

Model, $m = 3$					Reward			Runtime
n	States	Transitions	T'	T	$\mathbb{C}^A(\varphi_1)$	$\mathbb{C}^A(\varphi_2)$	$\mathbb{T}^A(\varphi_2, T')$	(seconds)
3	2,197	9,458	3	5	2.215	6.00	3.532	0.015
			10	10	6.371	6.381	6.381	
			30	15	6.384	6.381	6.381	
4	28,561	275,944	3	5	2.296	8.000	4.710	0.215
			10	10	8.575	8.598	8.598	
			30	15	8.582	8.598	8.598	
5	371,293	7,791,130	3	5	2.267	10.00	5.887	5.256
			10	10	10.763	10.810	10.810	
			30	15	10.770	10.810	10.810	

Since the target formula is **true** at exactly time T , this query computes the expected number of delivered items accumulated up to time T .

- The conditional expected accumulated reward until reaching *success* within time T , while avoiding *collide* states:

$$\mathbb{C}^A(\varphi_2) = \mathbb{C}^A(\neg \text{collide } \mathbf{U}^{[0,T]} \text{ success}).$$

- The conditional expected accumulated reward up to time T' on paths that reach *success* within time T , while avoiding *collide* states before that:

$$\mathbb{T}^A(\varphi_2, T') = \mathbb{T}^A(\neg \text{collide } \mathbf{U}^{[0,T]} \text{ success}, T').$$

Algorithms in PRISM and Storm can compute the first query, which is an unconditional expected accumulated reward over a fixed horizon, but existing methods cannot express the remaining conditional reward queries in this form. Our proposed extensions allow us to formally define all three queries and compute them efficiently. Table 2 reports the results. We consider $n \in \{3, 4, 5\}$ AGVs, each assigned a workload of $m = 3$ items. The resulting DMRMs have

$$(4 \cdot 3 + 1)^3 = 2,197, \quad (4 \cdot 3 + 1)^4 = 28,561, \quad (4 \cdot 3 + 1)^5 = 371,293$$

states for $n = 3, 4$, and 5 , respectively. The last column reports the runtime, in seconds, to compute the listed queries in our current implementation.

6 Related work

Extensive research has been published on rewards and conditional probabilities in Markov chains and Markov decision processes (MDPs), including logics like cpCTL [3] for conditional probabilities of path properties. [3, 10, 29] present methods to evaluate the conditional probability of reaching a state satisfying a given atomic proposition, conditioned on being on paths satisfying a linear temporal property. [10, 29] provide a prototype implementation in PRISM and compare the results with the probabilistic model checker STORM [1]. The primary focus of these papers is the computation of conditional probabilities; however, [29] also computes the expected time to reach a condition, though not

general expected rewards. [11] proposes algorithms to compute optimal conditional expected rewards under specific reachability goals in MDPs, whereas our work targets exact computation in DTMCs over bounded and unbounded until conditions with lower time bound $h \geq 0$. [20] computes the full reward distribution until a co-safe LTL formula is satisfied in DTMCs, but it neither handles $h > 0$ nor supports an observation time unrelated to the satisfaction time. More recently, [30] showed that conditional expected rewards conditioned on reaching specific absorbing states in a DTMC reduce to expected visiting time (EVT) computations, requiring a single linear system rather than one per goal; their setting is restricted to reachability conditions, with no path-formula structure and no time bounds, but shares with our work the goal of efficient conditional reward computation. Our extensions are complementary: we handle step-bounded until conditions with arbitrary lower bounds and introduce the $\mathbb{T}^A$, $\mathbb{T}^R$, and $\mathbb{T}^I$ operators that separate path-satisfaction time from reward-observation time.

Unlike [3, 10, 11, 20, 29, 30], which either focus strictly on probabilities and reachability properties or lack the ability to specify time-based reward accumulations under bounded-until conditions with a positive lower time bound, this paper introduces operators $\mathbb{C}^A$, $\mathbb{C}^R$, and $\mathbb{C}^I$ for conditional expected accumulated, rate, and impulse rewards, together with their time-focused counterparts $\mathbb{T}^A$, $\mathbb{T}^R$, and $\mathbb{T}^I$, and provides algorithms for their computation in DTMCs. Although the model-transformation approach of [10] can in principle be adapted to bounded-until conditions, to our knowledge no prior work provides a direct, explicit algorithm for conditional expected accumulated, rate, or impulse rewards under bounded-until conditions with a positive lower time bound. These extended reward operators allow more general performance and reliability queries of probabilistic systems, expanding PRCTL’s capabilities to handle complex queries that could previously neither be expressed nor computed.

PRISM [27] offers features to associate rewards with states or transitions, but provides algorithms to compute expected rewards only for unbounded until formulas, not for bounded until formulas. [26] extends CSL and proposes expected time and expected reward operators that compute the expected sojourn time the system spends in a state where the CSL formula is satisfied for a random time interval, or the expected reward before a random terminal time, and compare these quantities to a given threshold. This CSL extension could in principle be applied to PCTL, but its semantics are not comparable to the extensions proposed here, i.e., the two papers address different kinds of questions.

[23] proposes the computation tree measurement language (CTML), which allows $\{+, \times\}$ operations on until formulas and introduces a new operator to compute the expected reward to satisfy a path formula. The authors define state formulas that assign real values to the states of the DMRM, rather than the Boolean values used in PCTL. In CTML, an until formula with addition can be used to compute the expected reward to reach a goal state, but only when the lower time bound is $h = 0$; furthermore, when the upper bound is ∞ , the computation requires solving two systems of linear equations, whereas our approach requires solving only one.

Table 3: Relating formulas from [2] and [27] to our proposed logic.

Logic proposed in [2]	Logic proposed in this paper
For $t < \infty$, $\mathcal{E}_{\mathcal{R}}^t(\psi)$	$\mathbb{C}_{\mathcal{R}}^A(\varphi = \text{trueU}^{[t,t]}\text{true})$ with $\rho'[i] = \rho[i]/t$ if $i \models \psi$, $\rho'[i] = 0$ o.w.
For $t = \infty$, $\mathcal{E}_{\mathcal{R}}^\infty(\psi)$	$\mathbb{T}_{\mathcal{R}}^A(\text{trueUtrue}, \infty)$ with $\rho'[S]$
For $t < \infty$, $\mathcal{C}_{\mathcal{R}}^t(\psi)$	$\mathbb{C}_{\mathcal{R}}^R(\text{trueU}^{[t,t]}\text{true})$ with $\rho''[i] = \rho[i]$ if $i \models \psi$, $\rho''[i] = 0$ o.w.
For $t = \infty$, $\mathcal{C}_{\mathcal{R}}^\infty(\psi)$	$\mathbb{T}_{\mathcal{R}}^R(\text{trueUtrue}, \infty)$ with $\rho''[S]$
For $t < \infty$, $\mathcal{J}_{\mathcal{R}}^k(\psi)$	$\mathbb{C}_{\mathcal{R}}^A(\text{trueU}^{[t,t]}\text{true})$ with $\rho''[S]$
For $t = \infty$, $\mathcal{J}_{\mathcal{R}}^\infty(\psi)$	$\mathbb{T}_{\mathcal{R}}^A(\text{trueUtrue}, \infty)$ with $\rho''[S]$
Logic proposed in [27]	Logic proposed in this paper
For $t < \infty$, $\mathbb{R}_{\mathcal{R}}[I=t]$	$\mathbb{C}_{\mathcal{R}}^R(\text{trueU}^{[t,t]}\text{true})$
For $t < \infty$, $\mathbb{R}_{\mathcal{R}}[C \leq t]$	$\mathbb{C}_{\mathcal{R}}^A(\text{trueU}^{[t,t]}\text{true})$
For $t = \infty$, $\mathbb{R}_{\mathcal{R}}[C]$	$\mathbb{T}_{\mathcal{R}}^A(\text{trueUtrue}, \infty)$
$\mathbb{R}_{\mathcal{R}}[F\psi] \wedge \mathbb{P}(F\psi) = 1$	$\mathbb{C}_{\mathcal{R}}^A(\text{trueU}\psi) \wedge \mathbb{P}(F\psi) = 1$

Finally, to our knowledge, no prior work has addressed the computation of expected accumulated reward, expected reward rate, or expected impulse reward at a specific time t , conditioned on satisfying a path formula whose until operator has time bounds unrelated to t . This important extension allows for more sophisticated queries that distinguish, for example, between an absolute mission time and completion of individual mission objectives.

7 Conclusion and future work

We extended the probabilistic temporal logic PRCTL, defined on discrete-time Markov reward models, with new operators $\mathbb{C}^A$, $\mathbb{C}^R$, and $\mathbb{C}^I$. These compute the expected accumulated reward, reward rate, or reward impulse, conditioned on satisfying a path formula φ containing either a bounded, $\text{U}^{[h,k]}$, or unbounded, $\text{U}^{[h,\infty)}$, “until” operator, for an arbitrary lower bound $h \geq 0$ on the time interval. We also proposed operators $\mathbb{T}^A$, $\mathbb{T}^R$, $\mathbb{T}^I$ for expected accumulated reward up to time t , reward rate at time t , or reward impulse at time t , conditioned on satisfying φ , where t is unrelated to the time interval $[h,k]$ or $[h,\infty)$. To our knowledge, this separation of path-satisfaction time from reward-earning time is the first such treatment in probabilistic reward logic; this enables queries that neither PRCTL nor existing tools like PRISM and Storm can express. We proposed efficient “backward” algorithms to compute all these quantities, developed a prototype implementation in SMART [15] to verify system properties specified using our extended PRCTL, and evaluated it on two applications. In the future, we plan to define similar extensions for the probabilistic temporal logic CSL, to target the specification and verification of properties in continuous-time Markov reward models.

References

1. STORM: A tool for the analysis of systems involving random or probabilistic phenomena. <http://www.stormchecker.org/index.html>
2. Andova, S., Hermanns, H., Katoen, J.P.: Discrete-time rewards model-checked. In: Larsen, K.G., Niebert, P. (eds.) *Formal Modeling and Analysis of Timed Systems*. pp. 88–104. Springer Berlin Heidelberg (2004)
3. Andrés, M.E., van Rossum, P.: Conditional probabilities over probabilistic and nondeterministic systems. In: Ramakrishnan, C.R., Rehof, J. (eds.) *Tools and Algorithms for the Construction and Analysis of Systems*. pp. 157–172. Springer Berlin Heidelberg, Berlin, Heidelberg (2008)
4. Aziz, A., Singhal, V., Balarin, F., Brayton, R., Sangiovanni-Vincentelli, A.: It usually works: the temporal logic of stochastic systems. *LNCS* **939** (05 1995)
5. Baier, C., Cloth, L., Haverkort, B., Kuntz, M., Siegle, M.: Model checking action- and state-labelled Markov chains. In: *International Conference on Dependable Systems and Networks*, 2004. pp. 701–710 (2004)
6. Baier, C., Haverkort, B., Hermanns, H., Katoen, J.P.: Model checking continuous-time Markov chains by transient analysis. In: Emerson, E.A., Sistla, A.P. (eds.) *Computer Aided Verification*. pp. 358–372. Springer Berlin Heidelberg, Berlin, Heidelberg (2000)
7. Baier, C., Haverkort, B., Hermanns, H., Katoen, J.P.: On the logical characterisation of performability properties. In: Montanari, U., Rolim, J.D.P., Welzl, E. (eds.) *Automata, Languages and Programming*. pp. 780–792. Springer Berlin Heidelberg, Berlin, Heidelberg (2000)
8. Baier, C., Haverkort, B.R., Hermanns, H., Katoen, J.P.: Model checking meets performance evaluation. *SIGMETRICS Perform. Eval. Rev.* **32**(4), 10–15 (Mar 2005)
9. Baier, C., Katoen, J.P.: *Principles of Model Checking*. The MIT Press (2008)
10. Baier, C., Klein, J., Klüppelholz, S., Märcker, S.: Computing conditional probabilities in markovian models efficiently. In: Ábrahám, E., Havelund, K. (eds.) *Tools and Algorithms for the Construction and Analysis of Systems*. pp. 515–530. Springer Berlin Heidelberg, Berlin, Heidelberg (2014)
11. Baier, C., Klein, J., Klüppelholz, S., Wunderlich, S.: Maximizing the conditional expected reward for reaching the goal. In: Legay, A., Margaria, T. (eds.) *Tools and Algorithms for the Construction and Analysis of Systems*. pp. 269–285. Springer Berlin Heidelberg, Berlin, Heidelberg (2017)
12. Beaudry, M.D.: Performance-related reliability measures for computing systems. *IEEE Trans. Comput.* **C-27**(6), 540–547 (Jun 1978)
13. Bengtsson, J., Larsen, K., Larsson, F., Pettersson, P., Yi, W.: Uppaal—a tool suite for automatic verification of real-time systems. In: *Proceedings of the DIMACS/SYCON Workshop on Hybrid Systems III: Verification and Control: Verification and Control*. p. 232–243. Springer-Verlag, Berlin, Heidelberg (1996)
14. Ciardo, G., Jones, R., Miner, A.S., Siminiceanu, R.I.: Logic and stochastic modeling with SMART. *Performance Evaluation* **63**(6), 578–608 (2006)
15. Ciardo, G., Miner, A.S.: SMART: Stochastic Model-checking Analyzer for Reliability and Timing, available at <https://asminer.github.io/smart/>
16. Clarke, E.M., Emerson, E.A.: Design and synthesis of synchronization skeletons using branching-time temporal logic. In: *Logic of Programs*. pp. 52–71 (1981)
17. Clarke, E.M., Grumberg, O., Peled, D.A.: *Model Checking*. MIT Press (1999)

18. Dehnert, C., Junges, S., Katoen, J., Volk, M.: A Storm is coming: a modern probabilistic model checker. In: Proc. CAV 2017. LNCS, vol. 10427, pp. 592–600 (2017)
19. Donatelli, S., Haddad, S., Sproston, J.: Model checking timed and stochastic properties with CSL^{TA}. IEEE Transactions on Software Engineering **35**, 224–240 (March 2009)
20. Elsayed-Aly, I., Parker, D.: Distributional probabilistic model checking. In: NASA Formal Methods (NFM) (2024)
21. Hansson, H., Jonsson, B.: A logic for reasoning about time and reliability. Formal Aspects of Computing **6**, 102–111 (1994)
22. Itai, A., Rodeh, M.: Symmetry breaking in distributed networks. Information and Computation **88**(1), 60–87 (1990)
23. Jing, Y., Miner, A.S.: Computation tree measurement language (CTML). Form. Asp. Comput. **30**(3–4), 443–462 (2018)
24. Kemeny, J.G., Snell, J.L.: Finite Markov Chains. Van Nostrand (1960)
25. Kwiatkowska, M.: Model checking for probability and time: from theory to practice. In: Proceedings - Symposium on Logic in Computer Science. pp. 351–360 (07 2003)
26. Kwiatkowska, M., Norman, G., Pacheco, A.: Model checking expected time and expected reward formulae with random time bounds. Computers and Mathematics with Applications **51**(2), 305–316 (2006)
27. Kwiatkowska, M.Z., Norman, G., Parker, D.: Stochastic model checking. In: Formal Methods for the Design of Computer, Communication and Software Systems: Performance Evaluation. vol. 4486, pp. 220–270. Springer Berlin / Heidelberg (2007)
28. Kwiatkowska, M.Z., Norman, G., Parker, D.: PRISM 4.0: Verification of probabilistic real-time systems. In: Proceedings of the 23rd International Conference on Computer Aided Verification (CAV). pp. 585–591 (2011)
29. Märcker, S., Baier, C., Klein, J., Klüppelholz, S.: Computing conditional probabilities: Implementation and evaluation. In: Cimatti, A., Sirjani, M. (eds.) Software Engineering and Formal Methods - 15th International Conference, SEFM 2017, Trento, Italy, September 4–8, 2017, Proceedings. Lecture Notes in Computer Science, vol. 10469, pp. 349–366. Springer (2017). https://doi.org/10.1007/978-3-319-66197-1_22, https://doi.org/10.1007/978-3-319-66197-1_22
30. Mertens, H., Katoen, J.P., Quatmann, T., Winkler, T.: Accurately computing expected visiting times and stationary distributions in Markov chains. In: Tools and Algorithms for the Construction and Analysis of Systems (TACAS). LNCS (2024), arXiv:2401.10638
31. Tarjan, R.E.: Depth-first search and linear graph algorithms. SIAM J. Comput. **1**(2), 146–160 (1972)